

in turn controls a BLDC motor and the servo motors for rudder, aileron and elevator of the aircraft.

Arduino Uno board. Arduino is an open source electronics prototyping platform based on flexible, easy-to-use hardware and software. It is intended for artists, designers, hobbyists and anyone interested in creating interactive objects or environments.

Arduino UNO is a board based on ATmega328 microcontroller. It consists of 14 digital input/output pins, six analogue inputs, a USB connection for programming the on-board microcontroller, power jack, an ICSP header and a reset button. It is operated with a 16MHz crystal oscillator and contains everything needed to support the microcontroller. It is very easy to use as the user simply needs to connect it to a computer with a USB cable or power it with an AC-to-DC adaptor or battery to get started. The microcontroller on the board is programmed using Arduino programming language and Arduino development environment.

XBee module. XBee is a small 2.4GHz transceiver with a small microcontroller on board. It is a communication protocol which is used in many other instruments too.

There are two types of XBee radio transceivers: series 1 and series 2. Each series has two varieties: normal and pro. While the normal has a range of 30 to 40 metres in indoor or urban situations, along the line of sight it works up to 100 or even 125 metres. The pro is effective up to 1.6km on line of sight. However, with a Yagi directional antenna, XBee pro can reach as long as 20km on line of sight.

While series 1 has eight digital I/O pins, series 2 has 11 pins. Series 2 is more advanced and can be made into mesh networks whose range can be extended to any length by putting router XBees in between.

Analogue input and XBee

XBee analogue input pins read 0V to 1.2V maximum. Any voltage above 1.2V is treated as maximum or 1. Since most XBees operate at 3.3V, we reduce

this voltage to 1.2V by adding a suitable resistor along the input device.

Suppose the analogue sensor is an LDR whose impedance varies from 0 to 10k, to drop 3.3V to 1.2V across the LDR we put a 20k resistor in series with the LDR, so that two-third voltage (of 3.3V) is dropped before reaching the LDR.



Fig. 3: XBee module



Fig. 4: BLDC motor



Fig. 5: ESC module

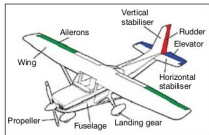


Fig. 6: Parts of RC aircraft (courtesy: dduino, blogspot.in)

Transmitter side. Pins 20 through 17 of XB1 are connected to four 10k presets (VR1 through VR4), which are used to control BLDC motor (refer Fig. 4), rudder, aileron and elevator, respectively.

Receiver side. Pins 0 and 1 of board1 are connected to pin 2 and 3 of XB2. Pins 9, 11, 10 and 12 of the board are connected to signal pins of M1 (elevator), M2 (rudder), M3 (aileron) servo motors and ESC module (refer Fig. 5), respectively. Main body parts of a typical RC aircraft are shown in Fig. 6.

BLDC motor is used for propeller of the aircraft. The propeller, or airscrew, converts rotary motion from motor to provide propulsive force. It is the most important part of the aircraft. The propeller is mounted on the front side of the aircraft and is generally made of two or more blades.

A rudder is used to steer the aircraft when it moves through the air, controlling the direction in which the aircraft is pointing. It is a flat sheet of material attached with hinges to the craft's stern, tail, or after end.

An aileron is a hinged flight control surface, usually attached to the trailing edge of each wing of an aircraft. Ailerons are used in pairs to control the aircraft in roll, or movement around the aircraft's longitudinal axis.

Elevators are flight control surfaces, usually at the rear end of an aircraft, which control the aircraft's longitudinal altitude. The position of the elevator controls whether the nose of the airplane is pointing up or down, and accordingly the plane moves up or down. The elevators are usually hinged to a fixed or adjustable rear surface.

Input supply wire of ESC is connected to 11.1V battery. Three output wires of ESC are connected to three-phase inputs of BLDC motor (M4).

Software

Arduino UNO is programmed using Arduino IDE software. ATmega328 on Arduino UNO comes with a boot loader that allows you to upload new